

When Grasp Metrics Disagree: Contact, Load, and Retention in Simulation

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Abstract—A simulated object can touch its environment without receiving load, and can receive load without depending on it for a specified continuation. Grasp evaluations that conflate these properties can disagree without either implementing its stated rule incorrectly. We separate literal no-contact, realized no-external-load, and intervention-conditioned retention, and test their consequences using frozen protocols in MuJoCo. Across 240 free-object controls, no-contact rejects 56 of 77 no-load reference positives; eight representation-only pairs change its verdict despite identical full-state trajectories. However, an integrated load-and-geometry monitor provides no observed gain over a full-wrench baseline on either registered primary comparison. It rescues none of 85 historical excluded scenes. On 120 fresh Allegro scenes, both monitors make identical selections or abstentions: 13 returns, 10 successful independent 15-second continuations, and 107 abstentions. Native-gate selection returns 40 candidates with 15 successful continuations, exposing a failure-versus-coverage tradeoff rather than uniform improvement. Fetch and Shadow adapters further reveal task and uncertainty boundaries. The result is a measurement study with explicit negative findings, not a superior grasp evaluator: evaluation targets, intervention horizons, and abstention costs must be stated before acceptance rates can support a claim.

Index Terms—Grasp evaluation, contact semantics, external load, retention, reproducibility.

I. INTRODUCTION AND CONTEXT

A grasping program lifts an object and holds it near the palm. Has it succeeded? A native task predicate may accept the motion; a continuous-hold predicate may reject an interruption; an environment-clear predicate may reject a contact record even when that record transmits no force. Conversely, measuring a nonzero environmental load does not establish that the grasp would fail if that load were removed. These are different questions about the same robotic execution, not interchangeable degrees of evaluator strictness.

This distinction matters when evaluation is used to select among candidate grasp schedules. A metric can better match a declared physical target yet reject useful candidates, or accept a historically unloaded trajectory that fails during subsequent use. The relevant outcome is then not just the fraction of returned candidates that work, but also failures and abstentions per assigned scene. Figure 1 grounds the distinction in free-object controls; the main decision experiment uses a dexterous Allegro hand and a fixed program library.

A. Relation to Grasp and Execution Evaluation

Large-scale grasp evaluation supports learned quality prediction and sim-to-real transfer, as in Get a Grip [1]. BODex

and task-oriented wrench estimation address grasp synthesis and mechanical capacity [2], [3]; Grasp to Act evaluates dynamic tool-use requirements [4]. These objectives differ from identifying loads that actually occurred along a recorded trajectory. External forces can also be intentionally useful, as established in extrinsic dexterity [5]; environmental contact is not intrinsically an error. SafeManip formalizes temporal safety properties, while Eval-Actions examines fine-grained execution quality [6], [7]. Our focus is narrower: when contact, realized load, and a specified removal test disagree, what has each measurement established, and does added monitor complexity improve a frozen selection decision?

B. Contributions and Scope

We contribute (1) a concrete separation of three evaluation targets, with full-state paired interventions and independently implemented reconstruction; (2) a frozen controlled study that establishes a representation-dependent contact mismatch but finds no incremental benefit of the integrated monitor over a strong full-wrench baseline; and (3) a sealed candidate-selection study that makes downstream use and abstention explicit. These contributions concern measurement validity and bounded negative evidence, not a new grasp planner, a learned evaluator, or a replacement for reinforcement learning.

Prior custom-hand and open-hand reports [8], [9] supply historical development context. The earlier compiled-physics study (V3) is reported separately in Section IV-E; the frozen measurement study (V4) supplies the primary comparisons. Their scene sets and evidential roles remain distinct.

II. EVALUATION TARGETS

A. Common Conditions and Literal No-Contact

For physical step t , let Q_t encode source-specific task conditions and G the unchanged episode-level eligibility rules. Allegro retains native success, settling and lift boundaries, thumb/opposing-region contact, and prohibitions on assistance or post-reset target-state edits. Its 10^{-4} N region-contact floor identifies nonzero contact, not support of the object’s weight. Let n_t^E count target–environment contact records, including inactive margin records. For continuous interval I and $T = 15$ s,

$$C_{\text{NC}} = G \wedge \exists I : |I| \geq T, \quad \forall t \in I, Q_t \wedge (n_t^E = 0). \quad (1)$$

This is a legitimate literal requirement. A rejection is a false negative only with respect to a *different, explicitly declared* reference target.

Comparison	Held fixed	Changed	Measured
Definition	Recorded trajectory	Contact / load rule	Paired verdicts
Intervention	Full state + input tape	Sham / removal	3 s retention cells
Selection	Same K candidates	First-pass choice	Sealed 15 s use

Fig. 2. Three distinct comparisons. Definition contrasts keep the trajectory fixed; branches change the physical continuation; selection changes the returned candidate. The separate 15 s future-use endpoint is evaluated only after choices are sealed.

TABLE I
FINAL UNITS AND FIXED SUBSETS. REFERENCE AND BRANCH SAMPLES ARE SUBSETS, NOT ADDITIONAL INDEPENDENT SCENES.

Source	Units	Task; branch / reference subset
Pad controls	240	18 s retention; 48 at 8 s / all 240
Allegro	120	Six families, 720 schedules; 20 rank-1 states / 30 rank-1 records
Fetch expert	100	Native 50 actions; separate 500-action retention; 20 / 30
Shadow / DGB	100	Four object identities; added gravity retention; 20 / 30

Table I lists final units. The controls use a free six-DOF target between actuated pads, without a target guide or hidden stabilizer. Their eight groups are hand-only, inactive-margin, necessary-floor, redundant-floor, opposed-contact, intermittent-floor, applied-assistance and boundary-clearance cases (30 each). Half the intermittent group has a single 0.1 s support burst; half has periodic support. Geometry, mass, friction and initial perturbations vary within locked ranges. Eight inactive-margin cases have a margin/gap-only partner; paired invariance is asserted only after full integration-state equality, not visual similarity.

The historical development retrospective covers every candidate slot in 85 V3 scenes accepted natively but not by no-contact. Of 510 slots, 177 qualifying traces received full-wrench replay; 333 already failed common conditions. Their unmeasured load histories are not imputed. This retrospective is not a fresh final test.

B. Comparators and External Boundaries

We use semantic names with frozen identifiers. Native (B0) is the original Allegro gate; on added control/external tasks it is an accumulated- Q proxy, not the upstream task score. Continuous (B1) requires contiguous Q ; no-contact (B2) is C_{NC} ; clearance (B3) adds positive geometric clearance. The primary full-wrench baseline (B4) uses the same force/torque thresholds, continuous time, passive-load accounting and uncertainty treatment as the integrated monitor (B5), but omits its geometry guard. B4 is not a weakened normal-force baseline. Normal-only (B4-normal) is reported separately.

Static feasibility (B6) uses an eight-ray friction cone and bounded actuator force balance; unsupported actuation is not applicable. A signed-force probe (B7) applies six one-weight world-axis forces for 1 s each. These measure capacity or perturbed future behavior, not historical no-load truth. They use the fixed branch subset and charge extra steps. Five no-load ablations remove geometry, torque, per-contact aggregation, uncertainty or continuous time; five control operating points were fixed before final outcomes.

Fetch uses the public scripted expert [12] with its native Gymnasium-Robotics goal/reset [13]; table-level goals are not raised. The added lift/two-finger-contact task retains target damping. Shadow/DexGraspBench (DGB) [14] begins from upstream pregrasp/grasp/squeeze, then adds gravity-on retention. It is not tabletop acquisition, and its environment-free model makes withdrawal a no-op. Allegro/Fetch use MuJoCo 3.8.0; DGB uses 3.3.2, not an independent engine. An intended native robomimic run was unavailable on this Windows host; DGB does not replace evidence from an independent learned acquisition program.

C. Sealed Selection and Primary Comparisons

All development criteria promote the same version from the genuine baseline/repair pool, so version-selection route U1 stops. Candidate route U2 keeps the repair library unchanged: select the first PASS in the fixed $K \in \{1, 3, 6\}$ prefix, or abstain. Every method observes and is charged all K schedules. Choices and score hashes are sealed before the future endpoint Y_{use} is generated. This endpoint is a separate 15 s continuation from the final state, with environmental contact removed and final commands held fixed, using the same all-sample pose/speed bounds. It does not read a monitor verdict. All 720 candidate futures are evaluated for finite-pool regret; decisions still have 120 scene units. Independence here means a separate outcome definition and sealed timing, not an independent dataset.

The registered H1 compares B5–B4 correct acceptance per assigned control subject to empirical accepted risk ≤ 0.05 . H4 compares usable returns per assigned scene at $K = 6$, retaining abstentions. The minimum practical gain is 0.05. Both use 10,000 paired scene bootstraps stratified by mechanism/family, seed 2026100101, with two-sided 97.5% Bonferroni-adjusted percentile intervals. Descriptive intervals are 95%; DGB resampling clusters by four identities. The budget neither guarantees power for five-point gains nor certifies low population risk. Zero-discordance intervals are not equivalence tests.

IV. RESULTS

A. Contact Representation and Realized Load

The controls contain 77 no-load reference positives and 163 negatives (Table II). Relative to that target, no-contact has no false acceptances but rejects 56/77 positives. In all eight registered margin/gap pairs, complete integration traces are identical while no-contact changes verdict; full-wrench and integrated monitoring do not. This demonstrates a representation effect in those pairs, not its frequency in natural grasps.

TABLE II
NOMINAL CONTROLS AGAINST THE DECLARED NO-LOAD REFERENCE.
FP/FN DENOMINATORS ARE 163/77. ALL ROWS HAVE ZERO
INDETERMINATE OUTCOMES.

Method	Returns	FP	FN	Accepted risk
Accumulated Q (B0)	201	124	0	61.7%
Continuous Q (B1)	182	105	0	57.7%
No-contact (B2)	21	0	56	0.0%
Clearance (B3)	77	0	0	0.0%
Normal-only (B4-normal)	77	0	0	0.0%
Full-wrench (B4)	77	0	0	0.0%
Integrated (B5)	77	0	0	0.0%
Accumulated no-load	92	15	0	16.3%

B. No Demonstrated Incremental Monitor Benefit

B4 and B5 both match all nominal control labels: H1’s observed difference is zero, with no paired discordance. Clearance and normal-only also match; these controls do not establish necessity of the integrated components. Accumulating unloaded time accepts 92 controls versus 77 for continuous time, with 15 extra acceptances in the single-burst group. Thus temporal continuity is consequential here, while the remaining nominal ablations do not show an incremental accuracy gain.

The fixed control branch subset contains 33 both-retain, six dependent and nine unreachable states. Redundant support can carry load while both branches retain the object. Load and dependence are therefore distinct under this specified test; empty four-cell categories are not filled with post-hoc cases. Natural branch reachability is limited: Allegro gives one dependent and 19 unreachable states; Fetch gives seven both-retain and 13 unreachable; DGB gives 16 both-retain and four unreachable, with no-op withdrawal. These counts cannot establish broad natural dependence prevalence.

C. No Historical Rescue; Source-Specific Outcomes

No-load rescues 0/85 historical excluded scenes. A controlled contact mismatch therefore does not explain away those failures. Native Fetch succeeds in 100/100 episodes, while the added retention task yields 0 definite passes, 52 failures and 48 indeterminate outcomes. Retained damping torque lies in the gray band; dropping torque would accept 48 but change the target. The native expert has not thereby failed its native task. DGB yields 66/100 native force-test successes versus 70 passes, 29 failures and one indeterminate outcome on added no-load retention. That indeterminate record also has invalid telemetry after a reproducible instability; it is not an extra 101st record. The 100 records span only four object identities.

Five of 720 Allegro records fail the frozen 10^{-10} m geometry-consistency assertion. A separately documented post-failure recovery retains them as invalid/indeterminate without widening the assertion, substituting scenes or exposing future-use scores. This recovery is not represented as pre-test code. Thirty probability-sampled natural references per source remain separate from control labels, with unknowns preserved.

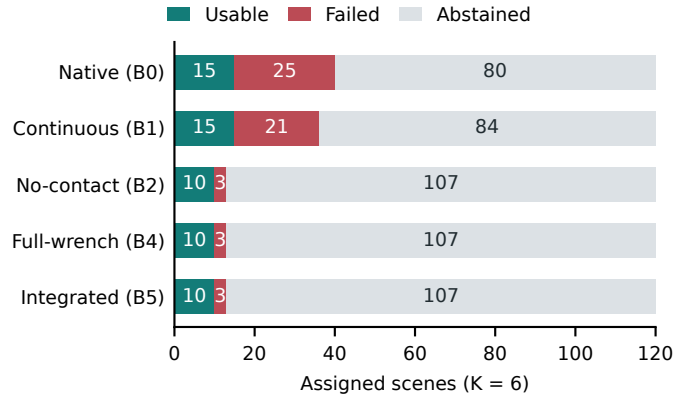


Fig. 3. K6 choices across all 120 scenes. Every returned candidate in these rows has a valid future-use outcome, so the middle segment denotes observed failure, not missing data. B2/B4/B5 choices are identical; B3 and B4-normal share these aggregate counts. All methods incur the same six-schedule budget.

D. Selection Trades Failures for Abstentions

At $K = 6$, B4 and B5 return the same candidate or abstention in all 120 scenes: 13 returns, 10 usable outcomes, and 107 abstentions. H4’s observed difference is zero. No-contact makes the same 120 choices. The zero bootstrap intervals for H1/H4 record this empirical identity, not population equivalence.

Native selection returns 40 candidates: 15 usable, 25 failed, 80 abstentions. Continuous selection returns 36: 15 usable, 21 failed, 84 abstentions (Fig. 3). Strict B2/B4/B5 selection yields 10 usable, three failed and 107 abstentions. Conditional usability rises from 15/40 (37.5%) to 10/13 (76.9%), but usability per assigned scene falls from 15/120 (12.5%) to 10/120 (8.3%). These are descriptive selection consequences, not a new confirmatory comparison. No utility winner follows without costs for a failed return and an abstention. Unknown futures elsewhere in the pool make its oracle a lower bound; the reported regret is not true optimal regret.

E. Historical Model and Library Context

The earlier V3 study addresses a different intervention: program repair under compiled primitive physics. On its fresh 360 scenes, K6 native/continuous/ no-contact successes were 128/122/37 for baseline and 141/133/48 for repair. All 11 no-contact gains were spheres, with no losses; removing the added topology anchors returned the baseline result. Repair’s no-contact counts were 14, 36 and 48 at $K = 1, 3, 6$. The gain was local to library content and budget, not broad dexterous grasping competence.

On the same earlier 120 scenes, recompiling the physical model changed no-contact counts from 18 to 13 for baseline and 24 to 18 for repair. Repair’s native total stayed 43, but seven cases gained and seven lost success. This changed geometry-dependent bounds and derived quantities as well as inertia; it was not an isolated inertia intervention. These paired historical results motivate explicit model and outcome definitions, but are not pooled with V4 or relabeled as its confirmatory monitor

benefit. Full tables and the original V3 uncertainty calculation are in the supplement.

V. PRACTICAL IMPLICATIONS AND LIMITS

A. Match Evidence to the Intended Claim

For literal separation, report contact semantics and margins. For absence of realized non-hand support, report force/torque paths, temporal intervals and unknowns. For retention after support removal, state the intervention, snapshot, controls and horizon. For selecting a grasp to use, report usable returns, failed returns and abstentions with a fixed candidate budget. A successful historical measurement cannot substitute for the corresponding future-use test. Nor is a stricter criterion inherently a better predictor.

The evidence supports using the simplest monitor adequate to the declared target. Here the full-wrench baseline already matches the integrated monitor on both primary comparisons. The observed geometry and torque redundancy on controls should not be generalized to all settings: Fetch torque changes uncertainty, and the fixed controls do not exhaust physically possible cases. B4/B5 use the same trajectories with no extra physics steps. Their recorded millisecond scoring times exclude instrumentation, I/O and simulation and do not establish a runtime advantage.

B. Validity and Access Boundaries

All evidence is simulated in one engine family, without hardware transfer. The hand library is finite; the control mechanism mix is constructed, not a population sample. Source tasks, reference sampling and branch reachability differ, so aggregate rates do not rank robots. Fixed open-loop removal omits controller adaptation, other removal mechanisms and longer horizons. Four DGB identities provide little support for broad object generalization. The positive control result, retrospective null and fresh selection null are all needed to avoid turning a diagnostic distinction into a capability claim.

The companion evidence package contains frozen protocols, individual scored records, sealed choices, future endpoints and offline aggregate reproduction. A separate core-raw companion supplies the final primitive-control traces; the full natural-task raw archive is not publicly hosted. Independent raw adjudication of the complete matrix and clean-machine dynamic reproduction are not claimed. Third-party dependencies retain their own licensing limits. The supplement states hashes, exact commands, recovery history, source versions, data-role boundaries and the limits of each verification level.

VI. CONCLUSION

Contact absence, realized load absence and retention after a specified removal are distinct evaluation targets. Frozen controls demonstrate that the first can disagree with the second without changing a trajectory, while redundant support separates loading from intervention-conditioned dependence. Yet the integrated monitor shows no gain over the full-wrench baseline, rescues no historical exclusions, and does not improve sealed downstream selection. The actionable result is to align the claimed property,

measurement and decision cost, and to retain null results and abstentions in the evidence.

DISCLOSURE

OpenAI Codex assisted code, analysis tooling and manuscript preparation under author direction. The author is responsible for all claims; no experiment isolates an effect of that assistance.

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